

Education

University of Illinois at Urbana-Champaign

Aug 2025 — May 2028

B.S. in Mathematics (Technical GPA: 4.0, Overall GPA: 3.98); Honors: James Scholar, Dean's List

- Relevant Coursework: System Programming, Computer Architecture, Data Structures, Abstract Algebra, Number Theory, Abstract Linear Algebra, Discrete Structures, Statistics & Probability, Differential Equations, Computational Science

Technical Skills and Awards

Awards: American Invitational Mathematics Exam (AIME) Qualifier, USACO Silver Division, Quiz Bowl Nationals

Languages: Lean4, Dafny, Python, C++, C, Verilog, Rust, SQL, React, TypeScript, HTML/CSS/JS

Tools & Libraries: Linux, Tensorflow, NumPy, MongoDB, FastAPI, AWS Bedrock, Claude Code/Codex/Cursor

Research

FOCAL Lab, Formal Verification & Verified LLM Decoding Research

Sept 2025 — Present

- Co-developed a verified constrained-decoding library in **Dafny** (1,900 lines) with **60+** cost-bounded CSD helpers. Pre/postconditions gate LLM-synthesized strategies before **Python** compilation.
- Integrated verify compile evaluate on **GSM-Symbolic**, **Spider**, and **SMILES**. Ran factorial studies across 4 HuggingFace evaluators (**Qwen** 1.5B/7B/14B, **Llama** 3.1 8B).
- Architected unified legacy adapters, stratified **100-example** splits, and syntax-annotated scoring. Compared **100+** configurations against CRANE, IterGen, CARS, and GCD to set **MetaDecode** stopping criteria.

Experience

AbbVie, Generative AI Platform & Workflow Automation Intern

June 2024 — Aug 2024

- Built a **Streamlit** UI enabling CRUD operations for **300+** JSON configuration files used by an internal Generative AI API and migrated **300+** prompt templates into a **PostgreSQL** database.
- Designed a modular configuration system enabling dynamic pipeline updates without code changes, reducing reconfiguration time from **hours to minutes** for internal AI workflows.

Technical Projects

Abstract Algebra, the Game, Interactive Lean4 Proof Game

- Developing an interactive **Lean4** proof game hosted on the Lean Game Server comprising **2 worlds** and **24 levels** that teach group theory through guided theorem proving, covering abelian groups, subgroups, and normal subgroups.
- Formalized **1,200+** lines of Lean4 encoding identity uniqueness, inverses, cancellation laws, commutators, conjugation, subgroup intersection, and the one-step subgroup test, building a notation-agnostic `MyGroup` typeclass.
- Designed progressive level sequences that decompose **formalized proofs** into **structured learning blocks**, with layered hints and narrative annotations guiding players from basic group axioms to advanced subgroup characterizations.

Two-Level Segregated Fit Malloc, High-Performance Memory Allocation

- Developed a custom memory allocator (malloc, calloc, realloc, free) in pure C, outperforming glibc by up to **100%** in throughput and achieving over **300%** memory reduction on select test cases, placing **3rd out of 300** students.
- Designed a **two-level segregated fit** (TLSF) allocator utilizing **128-bit** bitmask free-list indexing, localized arena allocation, and a deferred-free cache to optimize repeated alloc/free patterns.
- Engineered aggressive block-level optimizations, including compressed **32-bit** pointers, boundary-tag coalescing with footer elision, and flag-encoded headers for **O(1)** neighbor status verification.

Asynchronous Audio Relay, Low-Latency Concurrent Systems & Networking

- Architected a multithreaded UDP store-and-forward audio relay in C (Linux/Raspberry Pi) with **POSIX sockets** and per-peer **pthreads** to isolate connection state, replacing a single-threaded poll loop that serialized concurrent sessions and amplified head-of-line blocking.
- Engineered a **lock-free** ring buffer and append-only audio store for concurrent ingest and asynchronous voicemail playback, eliminating mutex contention on the hot path versus a mutex-backed prototype that introduced measurable tail latency under load.
- Validated **<500 ms** end-to-end latency on live two-party calls across a **900 mile** network path, demonstrating that store-and-forward relay semantics need not trade away interactive responsiveness.

Squishy.ai, *LLM-Driven Soft Robotics Simulation Platform*

- Built a full-stack platform that translates natural language into executable soft robotics physics simulations via PyElastica (Cosserat rod theory), placing **3rd** on the TRAE track among **200+** participants.
- Architected a high-performance **FastAPI** backend utilizing Numba JIT compilation, coupled with a headless Matplotlib and FFmpeg rendering pipeline for end-to-end simulation generation in **under 20 seconds**.
- Developed an interactive **React/Vite** frontend featuring real-time simulation visualization and LLM-routed code generation through the Keywords AI gateway.

SIGEcom, *Systematic Trading Strategy Research & Simulation*

- Designed, backtested, and deployed **multi-asset algorithmic trading strategies** in Python, utilizing SMA, EMA+RSI, MACD, and statistical arbitrage approaches across **6 symbols**.
- Integrated custom strategies into a live simulation framework, executing **40+ trades** over a one-week evaluation period on real-time cryptocurrency and FX market data.
- Achieved **positive PnL** across a diverse portfolio including BTC, ETH, SOL, INR, ZAR, and JPY.

Financial Engineering, *Quantitative Options Pricing & ML-Based Volatility Modeling*

- Implemented the **Cox-Ross-Rubinstein** binomial model in Python to price European and American options, verifying convergence toward Black-Scholes benchmark values.
- Leveraged **neural network de-Americanization** techniques to convert market American option prices into equivalent European prices, enabling **accurate implied volatility** computation.
- Analyzed **early exercise boundaries** and computational efficiency across various market regimes to optimize pricing accuracy.